

OLEN'YA RECHKA, Russian Federation

WMO: 299740

Lat: 52.8051N

Lon: 93.2406E

Elev: 4588

StdP: 12.42

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-14.0	-23.2	1.8	-17.3	-19.7	2.2	-13.0	31.3	-0.6	27.5	-1.8	6.3	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	73.1	57.5	69.9	56.6	66.9	55.2	60.1	68.6	58.5	67.0	56.8	64.8	5.9	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.0	82.1	63.0	55.1	76.5	61.3	53.5	72.2	59.8	28.7	68.7	27.5	67.5	26.3	65.0	66.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	19.7	17.5	DB	-25.3	80.3	8.1	2.7	-31.1	82.2	-35.9	83.8	-40.4	85.3	-46.3	87.2	(4)
(5)			WB	-25.6	63.6	8.0	1.8	-31.3	64.9	-36.0	66.0	-40.5	67.0	-46.4	68.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	0.6	4.9	18.1	29.2	38.5	51.3	55.4	52.9	42.5	27.9	11.0	5.4	(6)
(7)		DBStd	21.05	7.39	9.21	9.87	9.41	9.15	8.38	5.65	6.61	9.18	9.73	10.57	7.89	(7)
(8)		HDD50	8423	1532	1262	988	626	374	86	15	45	259	685	1169	1382	(8)
(9)		HDD65	13411	1997	1682	1453	1075	820	414	302	379	674	1149	1619	1847	(9)
(10)		CDD50	496	0	0	0	0	18	125	182	134	36	1	0	0	(10)
(11)		CDD65	10	0	0	0	0	0	3	4	3	0	0	0	0	(11)
(12)		CDH74	63	0	0	0	0	1	25	24	13	0	0	0	0	(12)
(13)		CDH80	2	0	0	0	0	0	1	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.6	10.0	7.5	4.9	4.3	4.0	3.7	3.3	3.5	4.1	4.8	7.5	9.7	(14)
(15)	Precipitation	PrecAvg	48.3	2.4	1.5	2.1	4.1	4.6	5.0	6.8	6.3	4.7	3.9	3.7	2.8	(15)
(16)		PrecMax	69.1	22.3	3.5	3.9	7.7	8.7	11.2	14.0	11.8	8.3	8.5	8.0	8.8	(16)
(17)		PrecMin	35.6	0.3	0.3	0.7	1.4	2.0	0.8	0.7	1.0	1.9	1.5	1.2	0.8	(17)
(18)		PrecStd	6.9	3.6	0.8	0.8	1.9	1.7	2.0	2.6	2.2	1.5	1.7	1.6	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	29.7	44.1	54.9	68.1	77.1	77.1	75.8	69.6	54.6	37.7	24.8	(19)
(20)			MCWB	16.8	27.5	34.7	40.7	49.3	58.8	61.3	58.5	52.5	41.3	32.1	23.4	(20)
(21)		2%	DB	15.7	24.2	39.5	50.2	62.1	72.6	72.7	71.1	64.4	49.5	31.9	20.1	(21)
(22)			MCWB	14.0	21.8	31.8	38.1	47.4	56.4	59.0	57.5	50.9	39.2	27.9	18.3	(22)
(23)		5%	DB	12.8	20.3	35.6	46.7	57.2	69.4	69.2	67.6	60.1	45.5	28.1	18.0	(23)
(24)			MCWB	11.4	17.7	29.2	36.4	44.9	55.6	57.8	55.6	49.5	37.5	24.5	16.5	(24)
(25)		10%	DB	9.8	17.0	31.6	43.0	52.9	65.7	66.0	64.2	55.8	41.4	24.2	15.7	(25)
(26)			MCWB	8.4	14.7	26.8	34.4	42.4	54.3	56.6	54.5	47.2	35.7	22.0	14.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	28.0	34.9	42.6	51.6	61.8	63.1	60.8	54.8	43.7	32.1	23.4	(27)
(28)			MCD B	17.7	29.0	42.3	52.4	64.5	70.9	72.6	69.5	65.0	49.6	36.3	24.3	(28)
(29)		2%	WB	14.1	22.1	32.4	39.3	48.3	59.0	60.7	58.9	52.3	40.6	28.0	18.7	(29)
(30)			MCD B	15.5	23.8	38.2	48.0	60.0	68.9	68.9	68.3	61.6	47.0	31.3	19.8	(30)
(31)		5%	WB	11.4	17.9	29.7	37.1	45.8	57.0	59.0	57.2	50.2	38.4	24.9	16.6	(31)
(32)			MCD B	12.8	19.9	35.2	44.5	55.2	66.6	66.9	65.8	58.7	44.7	27.7	17.9	(32)
(33)		10%	WB	8.5	14.7	26.7	35.3	43.4	55.4	57.4	55.3	47.8	35.9	22.2	14.4	(33)
(34)			MCD B	9.6	17.0	31.2	41.9	51.6	63.7	64.7	62.3	54.4	41.5	24.4	15.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	9.8	14.1	15.0	15.1	16.2	14.7	15.1	14.4	11.5	8.6	6.7	(35)
(36)			MCDBR	8.2	11.6	15.8	18.8	20.6	21.3	20.3	20.8	20.1	16.3	10.4	9.6	(36)
(37)			MCWBR	7.7	9.8	10.7	10.3	10.8	10.7	10.2	10.3	10.6	9.8	8.5	9.0	(37)
(38)		5% WB	MCD BR	8.1	11.6	15.1	16.6	18.9	18.8	16.9	18.6	18.4	14.8	9.3	9.3	(38)
(39)			MCWBR	7.8	10.1	10.7	9.7	10.9	10.3	9.5	9.8	10.2	9.3	7.9	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.285	0.327	0.366	0.374	0.362	0.365	0.352	0.323	0.312	0.285	0.256	(40)	
(41)		taud	2.098	2.067	2.030	2.131	2.241	2.388	2.403	2.417	2.450	2.330	2.122	2.037	(41)	
(42)		Ebn at Noon	229	256	268	273	277	282	279	277	273	247	210	198	(42)	
(43)		Edh at Noon	23	32	42	43	41	36	35	32	28	25	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	364	698	1153	1473	1683	1896	1719	1460	1059	620	326	240	(44)	
(45)		RadStd	28	44	64	80	134	178	120	117	111	56	38	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page